

# Where the S5 negative space lands

Follow-up: from an 800-query hole to a saturation measurement

*The first deck ended with the 800-query whole-skeleton negspace holding at zero across 5.4M structures, and the full 6,336-cell sweeps in flight. They are now complete. We then ran every fold-novelty probe we could reach — AFDB dark matter, TED novel folds, and a multi-filter annotation funnel — through the same S5 instrument. All land on known geometry. One probe looked like it broke the pattern; it turned out to be a methodological trap worth stating carefully, because it is the objection a skeptic will raise.*

# Recap — where deck 1 left off

- Enumeration:  $198 \text{ S5 skeletons} \times 32 \text{ H/E typings} = 6,336$  queries from the Chitturi 2016 2D-hex-lattice model.
- Whole-skeleton negspace: 25 skeletons (800 queries) return zero across every typing.
- Scale-up test:  $800 \text{ queries} \times (19\text{k curated} + 496\text{k experimental PDB} + 4.9\text{M AFDB}) = 2,400$  sweeps, 0 lit-up. Not one negspace motif is realized at 5.4M structures.
- Cost: four latent searchmatrix buffer/validation bugs, all fixed.
- Then in flight: the full 6,336-cell sweeps + the interactive inspector.

*Deck 2 picks up exactly here: the sweeps finished, and the question became how far does the negative generalize?*

# The full landscape is in — four databases, one matrix

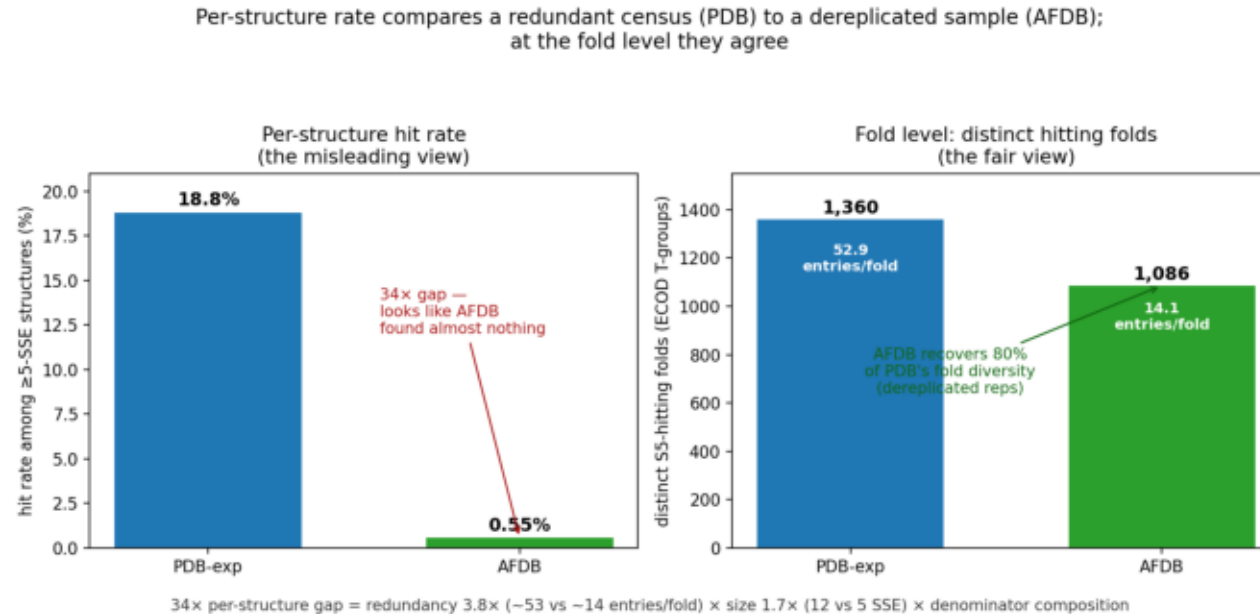
- 3,380 of 6,336 cells stay empty across all four (the per-cell negspace; a superset of the 800 whole-skeleton holes).
- All browsable in the inspector: heatmaps, per-cell exemplars rolled up by ECOD T-group, real per-pair contact geometry, all-pairs compositional enrichment.

# The reframe — the negative is the result

- S5 is an assignment-free instrument: pure SSE geometry, fully enumerable, so it comes with a denominator (6,336). That makes "what's missing" a measurable quantity, not an anecdote.
- A novelty test built on CATH/ECOD begs the question (novel = "we didn't classify it"). S5 does not.
- The claim: the observable 5-SSE fold repertoire is effectively closed. AlphaFold multiplies the membership of known folds by orders of magnitude but adds essentially no new local topology.

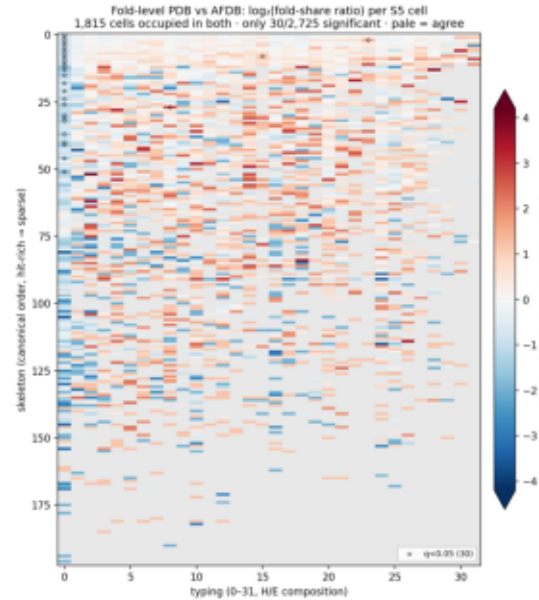
*Sequence space is exploding; it maps many-to-one onto a bounded, nearly-filled fold codomain that stopped growing orders of magnitude of sequence ago. AF is the lens that lets you see the collapse.*

# AFDB coverage: per-structure misleads, fold level agrees



- The trap: per-structure hit rate looks like AFDB found almost nothing — PDB-exp 18.8% vs AFDB 0.55% among  $\geq 5$ -SSE structures (34 $\times$ ). But that compares a redundant crystallization census to a dereplicated, smaller diversity sample — not a fair comparison.
- The fair view (fold level): distinct S5-hitting ECOD T-groups — PDB 1,360 vs AFDB 1,086. AFDB recovers 80% of the fold diversity all of experimental PDB does, from dereplicated reps.
- The 34 $\times$  decomposes: redundancy 3.8 $\times$  (PDB ~53 entries/fold vs AFDB ~14)  $\times$  size 1.7 $\times$  (AFDB search set is dereplicated cluster reps, and cluster-space is dominated by small-domain families  $\rightarrow$  median 5 vs 12 SSE)  $\times$  denominator composition (AFDB "capable" = the whole dereplicated proteome, mostly small/dark non-fold clusters). Not incompleteness — the sweep is complete (6,336 queries, 4.92M entries, verified).

# Cell by cell, PDB and AFDB agree at the fold level



- Redundancy-controlled diff:  $\log_2(\text{fold-share PDB} / \text{AFDB})$  per S5 cell, same skeleton $\times$ typing grid. Pale = agree.
- 1,815 cells occupied in both; only 30 of 2,725 differentially enriched ( $q < 0.05$ , circled). A faint PDB lean (it carries  $\sim 25\%$  more total folds) — but no cell class is AFDB's alone, and none PDB's alone at scale.
- Two independent samples of protein space — a redundant crystallographic census and a dereplicated predicted proteome — occupy the same motif cells in the same proportions. That convergence is the real evidence for a bounded repertoire.
- The AFDB-enriched cells are all HHHHH (all- $\alpha$ ) — and it's a confound, not new diversity: the same  $\alpha$ -solenoid/repeat families (ARM, Ankyrin, HEAT — X-group 109) + TM-helical (GPCR, MFS) tile overlapping 5-helix windows across many skeletons at once, so one family inflates every all- $\alpha$  cell. Composition (eukaryote repeats/TM)  $\times$  S5-blind-to-helical-identity  $\times$  solenoid tiling — not reachable topology.

# Why the fold repertoire looks closed — and the honest caveat

- The fold-level agreement (AFDB  $\approx$  80% of PDB's S5-hitting folds, from a totally
- AFDB's lower count (1,086 vs 1,360) is shallower sampling of the rarer folds,
- Caveat kept in view: "fold" here = ECOD T-group as the dereplication unit;

*independent, dereplicated sample) is the positive statement: the two views of protein space converge on the same bounded set of local motifs.*

*not a different repertoire — it adds essentially no cells PDB lacks.*

*the per-structure rate is genuinely uninformative across datasets of different redundancy + size, so all cross-dataset claims are made at the fold level with the completeness audit attached.*

# The novelty gauntlet — every probe we could reach

- Four independent definitions of novelty — geometric, dark-sequence, no-CATH, and annotation-filtered.
- All land on known S5 geometry. The one apparent exception (v3) is a methodological artifact — next slides.

*Each pushes a differently-defined "new fold" set through the same S5 instrument.*



# Probes 1-2 — dark matter and TED novel folds

- ~16% of the DB but only ~4% of hits — it under-hits S5.
- 786 occupied cells, 96% already PDB-known; ~16 novel-vs-all-known, all singletons.
- 89% land on PDB-occupied cells.
- Exactly 1 robust (multi-structure) novel cell: skeleton 118 / EEHHE.
- Non-circular: S5 knows nothing about CATH, yet agrees the geometry is known.

*AFDB dark matter (sequence-dark  $\neq$  fold-dark):*

*TED novel folds (defined by no CATH match — a classification-based novelty claim):*

*Even the most-curated fold-novelty set on the planet is, at the 5-SSE level, geometry we have already seen.*

## Probe 3 — the annotation funnel, and the trap

- `~/work/afdb_200M` GENUINE funnel: dark clusters → no-Pfam → no-CATH → no-InterPro →  $pLDDT \geq 70$  → low-decay → not-prophage →  $\geq 10$  members. 4,201 "genuine novelty" candidate domains.
- Searched as whole multi-domain models, they appear to hit 40 robust never-seen cells (top: HEEHE, 6 independent proteins) — the only probe to break the pattern.
- The catch: those motifs draw their five SSEs from scattered positions across a whole protein (median span 234 res vs 153 for known-cell hits) — i.e. across domain boundaries. An arrangement a domain-chopped reference search structurally cannot produce.

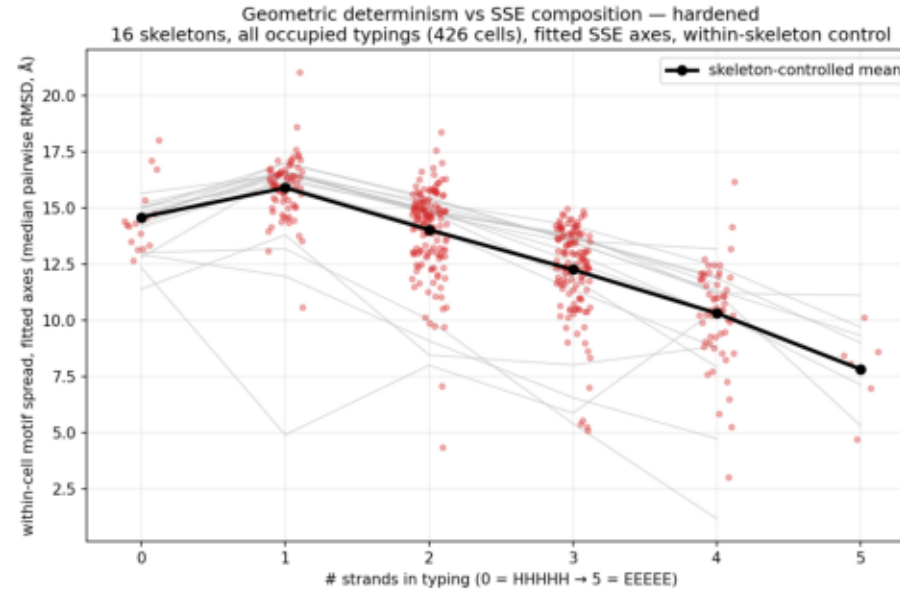
*The reference databases were searched as chopped domains; v3 was searched as whole models. That granularity mismatch — not novelty — manufactures never-seen cells.*

# The sealing test — a controlled A/B

- The 40 robust "novel" cells  $\rightarrow$  0. The 25 residual never-seen cells are all singletons (noise floor).
- Lesson for the paper: whole-model vs domain-chopped input must be matched, or cross-domain SSE assembly masquerades as fold novelty.

*Chop each candidate to its own committed  $_D<n>$  domain boundary (the pipeline's own decomposition — no arbitrary cutoff), rebuild with the identical PALSSE + generateMatrix pipeline. Only one variable changes: chopping.*

# When to trust S5 — the instrument's own limits



- Composition-graded fidelity: within-cell 5-SSE geometry tightens monotonically with strand count ( $\sim -1.9$  Å/strand); a 1-strand penalty — all- $\alpha$  / single-strand cells are grab-bags,  $\beta$ -registration makes a cell determinate.
- It's baked into the query, not just the structures: an EEEEE query carries  $\sim 700$  hard - (non-contact) constraints +  $\sim 1,000$  sheet-pairing codes (c/t) across the 198 skeletons; HHHHH is  $\sim 854$  X wildcards with zero hard constraints. Same-sheet non-adjacent strands cannot touch (register-separated)  $\rightarrow -$ ; non-adjacent helices are unconstrained  $\rightarrow X$ . So all- $\alpha$  cells are intrinsically looser queries and match promiscuously — the fidelity gradient is half structural, half encoding.
- Blind to transmembrane: GPCR (5001.1.1) and MFS (5050.1.1) — maximally different TM folds — both land in generic HHHHH cells.
- Tiling inflates abundance: ARM repeats (109 4 1) — one domain hits up to 123 cells. Abundant  $\neq$  novel:

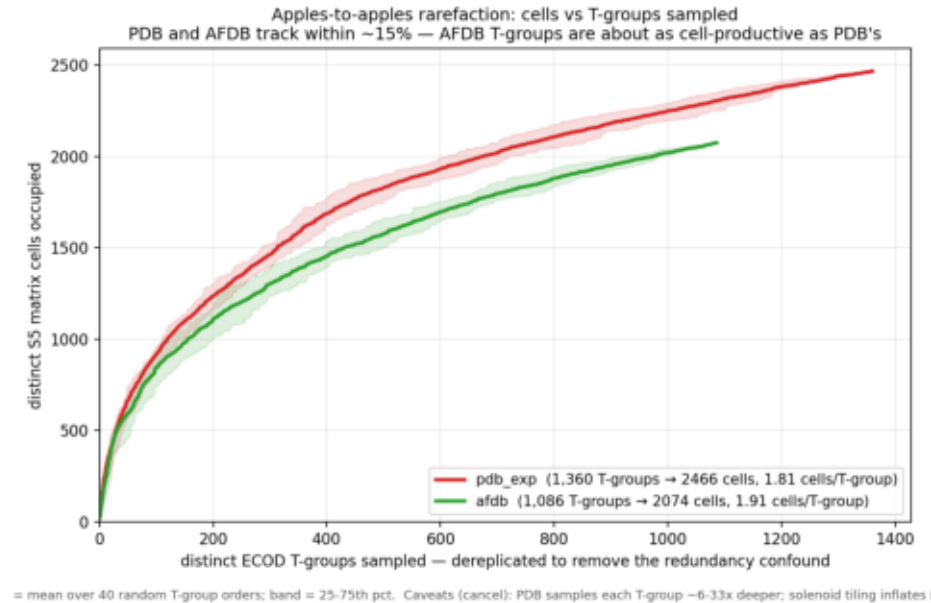
# Where it lands

- "ECOD is completed" — bounded to three regimes: (1) the common core is discovery-complete; (2) the frontier moved from discovery to membership (AF = remote-homology-at-scale); (3) the tail is curation-hard, not discovery-rich (repeats, symmetry, ARM).
- Bottom line: AFDB does not robustly reach S5 cells PDB can't — it's a ~80% subset of PDB's folds sampled far more densely, and every novelty probe lands on known geometry. Two caveats bound this:
  - Caveat 1 — S5 is a local, fixed-cardinality descriptor (Lesk): it measures 5-SSE local motif geometry — blind to global topology, transmembrane arrangement, domain-scale novelty. "No new S5 cells"  $\neq$  "no novel global folds"; an empty cell = "never observed here," not "unfoldable."
  - Caveat 2 — AlphaFold was trained on the PDB: AFDB agreeing with PDB on what's reachable is partly a training prior. The data can't cleanly separate "the real proteome is saturated" from "AF only renders PDB-like geometry." Same practical result (broader prediction surfaces no new cells), genuinely ambiguous mechanism.
- Still open: are the 3,380 empty cells sterically reachable-but-untaken, or lattice over-generation? Resolution needs constructive 3D backbone modeling (RFdiffusion / Rosetta) on the negspace.

# Recap

- Full landscape done: 6,336 cells × 4 databases; 3,380 stay empty; all in the interactive inspector.
- The negative reframed: a saturation measurement with an assignment-free, enumerable instrument — the observable 5-SSE repertoire is effectively closed.
- PDB  $\approx$  AFDB at the fold level: per-structure hit rate misleads (34×, redundancy×size); dereplicated by fold, AFDB recovers 80% of PDB's S5-hitting folds (1,086 vs 1,360) — coverage is redundancy, not incompleteness.
- Novelty gauntlet: four independent novelty definitions (lattice, dark-sequence, no-CATH, annotation-funnel) → all land on known geometry.
- The one trap: whole-model search fabricated 40 "novel" cells; a controlled domain-chopped re-search collapsed them to 0. Input granularity, not topology.
- Deliverables: inspector (leda:3001/docs), sealed capstone, full session log.
- Open: geometric reachability of the 3,380 — constructive modeling, not more search.

# Backup — apples-to-apples rarefaction (per T-group)



- The honest replacement for the retired per-structure rarefaction: x-axis is distinct ECOD T-groups, not structures → removes the redundancy confound (PDB census vs AFDB dereplicated).
- PDB and AFDB track within ~15% — AFDB T-groups are about as cell-productive as PDB's (1.91 vs 1.81 cells/T-group). No dataset samples matrix-space dramatically better per T-group.
- Band = IQR over 40 random T-group orderings (tight → order-robust). Residual caveats partially cancel: PDB samples each T-group ~6-33x deeper (union → more cells); solenoid tiling inflates cells/T-group in both.